Real-time keyword spotting on ARM Cortex-M4F

Build and test

Continuous single-keyword spotting on an **Arduino Nano 33 BLE / nRF52840** (64 MHz Cortex-M4F). Interrupt-driven PDM acquisition feeds CMSIS-DSP feature extraction and an int8 TensorFlow Lite Micro model in a dedicated audio task. A second task filters queued scores and controls the device. Confirmed **"yes"** detections toggle the built-in LED. The application runs continuously while powered.

The implementation focuses on bounded buffering, preserved sample/frame order, measured execution time, explicit module ownership, and tested failure paths. The primary workflow is command-line Make and C++; Arduino IDE is optional.

Measured results

Measurements below are from the two-task RTOS firmware on the connected Nano, recorded October 20, 2026. Timing summarizes normal-operation checks; the deliberate stall is documented separately. Memory is from the locked CLI build. These are observed results, not worst-case guarantees.

Measurement	Recorded result
Audio rate / hop	16 kHz / 256 samples (16 ms)
FFT frame / overlap	512 samples (32 ms) / 50%
Model input	61 x 20 int8 log-mel features (~1 s context)
Decision cadence after warmup	Every 8 feature hops (~128 ms), when audio backlog is drained
DSP average / observed maximum	2.63 / 2.86 ms
Inference average / observed maximum	59.13 / 59.65 ms
Processing iteration observed maximum	62.23 ms
Circular audio buffer / observed peak occupancy	8192 / 768 samples
Flash / static RAM	331,440 / 209,320 bytes (33% / 79%)
Dedicated task stacks	Audio 8192 bytes; control 6144 bytes
RTOS spoken "yes" check	11 confirmed detections; physical LED toggling confirmed
Reported sample drops / missed buffers / inference errors	0 / 0 / 0 in the recorded checks

See RTOS validation for current task, stress, and speech-test evidence. The earlier module/refactor record and pre-RTOS build check retain historical results. Response measurements start at a high-confidence window's estimated end; they do not measure acoustic word-onset latency. Detection accuracy across a controlled speaker/noise dataset and hours-to-days endurance have not been established.

Build, test, and flash

Prerequisites: **Python 3.9+**, GNU Make, and a host C++ compiler with AddressSanitizer/UndefinedBehaviorSanitizer support. Supported firmware-build hosts: macOS Intel/Apple Silicon and Linux x86-64/ARM64. Network access is needed for initial setup. Apple Silicon also needs Rosetta 2 for the pinned Intel Arm compiler. No Arduino IDE or globally installed Arduino CLI is required.

```
git clone https://github.com/jameslavrenz/KWS_Arm_CortexM4F.git
cd KWS_Arm_CortexM4F
make setup                        # Download and verify pinned tools/libraries
make test                         # Host unit and integration tests
make build                        # Cross-compile for Nano 33 BLE
make boards                       # Find the connected board's serial port
make flash PORT=/dev/cu.usbmodem11201  # Example macOS port; use your board's port
# Linux example: make flash PORT=/dev/ttyACM0
```

`make build` writes ELF/BIN/HEX outputs and a source/dependency manifest to `.build/artifacts/`. `make flash` rebuilds before uploading. All downloaded tools, board packages, libraries, generated sketch files, and objects stay under the ignored `.build/` directory. Existing Arduino IDE libraries are not used. Use `JOBS=4` to change firmware compile parallelism. A board is needed only for `make boards`, flashing, and physical tests; host tests and compilation can run without one. `make help` lists the available targets.

For prerequisite installation, expected output files, serial-port examples, troubleshooting, and rebuilding the PDF guide, follow the complete build guide.

dependencies.lock.json pins Arduino CLI 1.5.1, Arduino mbed Nano core 4.6.0 (GCC Arm Embedded 7-2017q4), CMSIS-DSP 5.7.0, and TensorFlow Lite Micro's exact source revision. CLI/library downloads are SHA-256 checked; Arduino CLI verifies core/tool package downloads. The three required TFLite compatibility headers are tracked in `build-support/`. See build reproducibility for dependency provenance and limitations.

GitHub Actions runs **the same** `make test` **and** `make setup && make build` **paths** on pushes and pull requests. Firmware and build manifests are retained as CI artifacts. Physical microphone/LED testing remains a hardware check.

Runtime architecture and ownership

```
PDM / DMA -> interrupt -> circular audio buffer
Audio task: frame overlap / FFT / log-mel -> int8 model
  -> eight-entry RTOS value queue -> control task
Control task: confidence filter -> LED; USB telemetry
```

Module	Owned state and responsibility	Execution context
`audio_capture`	PDM staging, ring, acquisition counters, complete-hop reads	Interrupt producer; audio-task consumer
`audio_pipeline`	Hop processing, readiness, backlog deferral, DSP/inference timing	Audio task
`feature_extractor`	Hann/FFT workspace, overlap, quantization, rolling features	Audio task
`keyword_model`	Interpreter, 96 KiB tensor arena, class-0 probability	Audio task
`keyword_event` / `keyword_detector`	Event freshness, confidence filtering, cooldown, response evidence	Control task
`device_control`	Built-in LED state and confirmed-event action	Control task after initialization
`telemetry`	Output staging and nonblocking USB transmission	Control task after initialization

Module	Owned state and responsibility	Execution context
`kws_pipeline`	Task/queue startup, transport, generations, metrics synchronization	Startup plus two task entry points
`kws_config`	Shared dimensions and inference cadence	Compile-time configuration

The `.ino` only delegates startup and idles the existing Arduino main task. Two application tasks use the board's built-in CMSIS-RTOS2/RTX kernel: audio at `AboveNormal` priority with an 8 KiB stack, control at `Normal` with a 6 KiB stack. Stacks, thread control blocks, queue storage, and mutex storage are static. The callback sets an ISR-safe thread flag after publishing audio. Flags may coalesce; the audio task drains complete hops before waiting again.

The queue copies timestamped score values, including low scores needed to rearm the detector. No audio buffers, feature windows, or tensor pointers cross tasks. A priority-inheriting mutex protects only a short metrics copy. The interrupt callback never takes that mutex. Audio never waits for room in the event queue. See RTOS design for scheduling, ownership, and overload rules.

Real-time behavior and design choices

Acquisition. The PDM callback drains bounded reads into a single-producer, single-consumer circular buffer using lock-free atomic publication. It does no DSP, inference, GPIO action, or USB printing. The audio task consumes complete 256-sample hops. The 8192-sample capacity provides 512 ms of nominal buffering at 16 kHz; it absorbs bounded stalls rather than sustained overload.

Scheduling. Capture starts after model/DSP initialization. The first decision waits for a full feature window; subsequent decisions require eight new feature frames and less than one queued hop. FFT frame overlap continues across model calls. DSP and inference leave interrupts enabled; short counter snapshots mask interrupts. With the recorded averages, eight DSP hops plus one inference take about 79 ms of a nominal 128 ms decision interval, excluding other work.

FFT correctness. CMSIS fast real FFT packs DC at output index 0 and Nyquist at index 1. `fft_power.h` treats both separately and computes interior-bin power without reading beyond the 512-float output. Tests exercise exact-size arrays under address/undefined-behavior sanitizers.

Detection. Class 0 is `yes`; the training labels also include unknown speech and silence. A confidence EMA with weight 0.5 must reach 0.75 while raw confidence is at least 0.8 on two successive decisions. The filter rearms below 0.4 and requires 1500 ms between triggers. Thresholds are starting values for broader speech/noise validation, rather than an established false-positive guarantee.

Failure behavior. A full ring rejects new samples, counts dropped samples and affected callbacks, and preserves unread samples. On observing a loss, the audio task discards pending audio, resets features, invalidates older queued scores, and waits for a fresh feature window. Failed inference increments an error counter and invalidates detection evidence. A full event queue drops a score and advances a generation counter rather than blocking audio. The control task rejects old generations and scores older than 250 ms; sequence/cadence gaps reset temporal evidence while retaining latch/cooldown history. Startup checks DSP/model/PDM and RTOS objects/tasks, cleaning up tasks on partial failure. Automatic recovery from peripheral failure is future work.

Tests and instrumentation

`make test` exercises queue ordering/overflow/wraparound, packed FFT bins/bounds, simulated inference stalls, temporal filtering and cooldown, detection-response timing and timer wraparound, actual GPIO-module logic with fake pins, and the actual audio pipeline with fake capture/features/model modules. RTOS startup tests check static storage, priority/stack configuration, callback notification, and cleanup after initialization failures. Event tests cover stale generations, queue gaps, duplicate scores, timestamp wraparound, and cooldown retention. Build-tool tests check archive checksums, extraction boundaries, library/lock drift, and environment isolation.

At 115200 baud, USB telemetry is emitted every two seconds without waiting for a host. A two-second startup timeout allows standalone operation.

Field	Meaning
`samples`, `dropped`, `missed_buffers`	Received sample count; ring loss; callbacks with loss
`queue_max`, `gap_max_us`, `audio_age_us`	Peak backlog; maximum callback gap; time since last callback
`dsp_us`, `infer_us`, `process_us`	Average/maximum execution times in microseconds
`response_us`	Last/maximum estimated high-confidence-window-to-action delay
`inferences`, `errors`, `p_milli`	Model calls, failures, class-0 confidence x 1000
`triggers`, `led`	Confirmed actions and current logical LED state
`event_drops`, `stale_events`, `notify_errors`	Nonblocking queue send failures, discarded stale scores, ISR flag errors
`event_us`	Average/maximum inference-result transport delay
`stack_free`	Minimum unused stack bytes: audio/control, from the RTX watermark

For a live capture, install `pyserial` in a host virtual environment and run:

```
python3 tests/hardware_capture.py --port /dev/cu.usbmodem11201 --seconds 60
```

Say "yes" with pauses of about three seconds and observe the LED. The script checks acquisition/inference progression and reported loss/error counters; physical LED behavior and speech-trigger correspondence require observation. The host recording ends after the requested interval; firmware keeps running.

Model and project artifacts

- Training notebook: Google Speech Commands training and quantization workflow.

- Model architecture: exported model visualization.

- Feature timing diagram: context and overlap illustration.

- Documentation index: initial and refactor validation records with raw telemetry.

Next engineering work

Evaluate false accepts/rejects across speakers, noise, distance, and silence; run longer endurance tests; add peripheral fault recovery; and extend the control task with communications or additional device functionality. Threshold refinement and multi-keyword training are separate from the build/runtime work.